BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 2481 / 2488

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

Q. 1123 (Inter Part - I) — Chemistry (Objective) — Session 2019-21 to 2022-24 — (Group - I) — Paper (I)

Q. Time Allowed:- 20 minutes — Maximum Marks:- 17 — PAPER CODE 2481

Note:- You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of Ink Remover or white correcting fluid is not allowed.

Q. Q. 1

Q1. The volume occupied by 1.4 g of N_2 at S.T.P. is

(A) (B)

(C) (D)

Q2. Which of the following is a monoisotopic element.

(A) (B)

(C) (D)

Q3. Which of the following can be sublime.

(A) (B)

(C) (D)

Q4. Constant factor in charlie's law.

(A) (B)

(C) (D)

Q5. The order of rate of diffusion of gases NH_3, SO_2, Cl_2 and CO_2 is

(A) (B)

(C) (D)

Q6. Which of the following is amorphus solid

(A) (B)

(C) (D)

Q7. Which of the following has highest vapour pressure at 25°C.

(A) (B)

(C) (D)

Q8. When 6d orbital is complete the entering electron goes into

(A) (B)

(C) (D)

Q9. Number of bonds in nitrogen molecule is

(A) (B)

(C) (D)

Q10. Units of energy in which heat changes in S.I system are.

(A) (B)

(C) (D)

Q11. The net heat change in a chemical reaction is same weather the reaction completes in one step or several steps. It is known as

(A) (B)

(C) (D)

Q12. Mixture of NH_4OH and NH_4Cl makes a buffer whose pH is

(A) (B)

(C) (D)

Q13. For the reaction $N_2+3H_2 \rightleftharpoons 2NH_3$, The pressure at optimum condition is.

(A) (B)

(C) (D)

Q14. Molarity of pure water is.

(A) (B)

(C) (D)

Q15. If a strip of Cu metal is placed in a solution of $FeSO_4$

(A) (B)

(C) (D)

Q16. Oxidation number of Mn in $KMnO_4$ is

(A) (B)

(C) (D)

Q17. The unit of rate constant is the same as that of the rate of reaction in

(A) (B)

(C) (D)

> Note: Board typos kept verbatim: 'charlie's law', 'Tempreture', 'amorphus', 'weather', "Hesse's law".

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

Q. 1123 (Inter Part - I) — Chemistry (Subjective) — Session 2019-21 to 2022-24 — Group (I) — Paper (I)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) N_2 and CO have the same number of electrons, protons and neutrons. (ii) 'Mg' atom is twice heavier than that of carbon atom. (iii) How can the efficiency of a chemical reaction can be expressed? (iv) List the four postulates of Kinetic molecular theory of gases. (v) What are characteristics of plasma? (vi) Throw some Light on the factor 1/273 in charle's Law. (vii) The e/m value of positive rays for different gases are different but those for cathode rays the e/m values is the same. Justify it. (viii) What are the defects of Bohr's atomic model. (ix) Compare line emission and line absorption spectra. (x) What is a spontaneous process? Give examples (xi) Why is it necessary to mention the physical states of reactant and products in a thermochemical equation? (xii) Define state and state function's with one example for each.

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) What is parts per million. Write its formula? (ii) What are the conditions should be fullfiled to observe colligative properties. (iii) Define hydrates. Give example. (iv) What is

activation of catalyst. Give one example? (v) How surface area has effect on the rate of reaction? (vi) Catalyst are specific in their action. (vii) Why sintered glass crucible is better than gouch crucible? (viii) Write down major steps involved in complete quantitative analysis. (ix) How mixture of sand and naphthalene can be separated? (x) Earthenware vessel keep water cool. Justify. (xi) Define symmetry. What are symmetry elements. (xii) Ionic solids are highly brittle in nature.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) Define Bond Energy? (ii) A Salt Bridge maintains the electrical neutrality in the cell. Justify it. (iii) Why cationic radius is smaller than atomic radius? (iv) Why 2nd Ionization Energy is always greater than first Ionization Energy? (v) What is pKb? Give its significance. (vi) Define pH? (vii) What does mean by chemical Equilibrium? (viii) What is oxidation number? Give example. (ix) Define Electrolysis.

Q. (Marks 8 × 3 = 24) Section -------- II Note: Attempt any three questions.

Q5. 5. (a) Describe combustion analysis for the determination of percentage of C, H and O in an organic compound.

(b) Calculate the mass of 1 dm^3 of NH_3 gas at 30°C and 1000 mm Hg pressure, considering that NH_3 is behaving ideally.

Q6. 6. (a) Describe Manometric method for determination of vapour pressure of a liquid with a diagram.

(b) What is Enthalpy of a reaction? How ΔH of a reaction is measured in Laboratory by glass calorimeter?

Q7. 7. (a) Explain Heisenberg uncertainity principle.

(b) The solubility product of Ag_2CrO_4 is $2.6x10^{-2}$ at 25°C. Calculate the solubility of compound. Atomic mass of Ag=108 Cr=52 O=16.

Q8. 8. (a) What is orbital hybridization? Explain the structure of CH_4 molecule on the basis of hybridization theory.

(b) Describe the construction and working of standard hydrogen electrode (SHE).

Q9. 9. (a) Explain continuous and discontinuous solubility curves. (b) Describe energy of activation in detail.

> Note: Board typos kept verbatim: 'charle's Law', 'fullfiled', 'gouch', 'uncertainity'.

GROUP – II

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

Q. 1123 (Inter Part - I) — Chemistry (Objective) — Session 2019-21 to 2022-24 — (Group - II) — Paper (I)

Q. Time Allowed:- 20 minutes — Maximum Marks:- 17 — PAPER CODE 2488

Note:- You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of Ink Remover or white correcting fluid is not allowed.

Q. Q. 1

Q1. Which of the following has hydrogen bonding?

(A) (B)

(C) (D)

Q2. The electron affinity of chlorine is.

(A) (B)

(C) (D)

Q3. Acid having Ka > 1 will be .

(A) (B)

(C) (D)

Q4. 18 g glucose is dissolved in 90 g of water. The relative lowering of vapour pressure is equal to

(A) (B)

(C) (D)

Q5. Orbitals having same energy are called:

(A) (B)

(C) (D)

Q6. The volume of 1.6g of CH_4 at S.T.P is

(A) (B)

(C) (D)

Q7. Partial pressure of oxygen in air at sea level is.

(A) (B)

(C) (D)

Q8. In silver oxide battery, the cathode is made up of:

(A) (B)

(C) (D)

Q9. For the reaction $NaOH + HCl \rightarrow NaCl + H_2O$ the change in enthalpy is called:

(A) (B)

(C) (D)

Q10. Stronger the oxidizing agent, greater is the:

(A) (B)

(C) (D)

Q11. The rate of reaction.

(A) (B)

(C) (D)

Q12. The largest number of molecules are present in:

(A) (B)

(C) (D)

Q13. Solvent extraction method is a particularly useful technique for separation when the product to be separated is.

(A) (B)

(C) (D)

Q14. The order of the rate of diffusion of gases NH_3, SO_2, Cl_2 and CO_2 is:

(A) (B)

(C) (D)

Q15. In order to raise the boiling point of water upto 110°C, the external pressure should be

(A) (B)

(C) (D)

Q16. Which of the following molecules has zero dipole moment?

(A) (B)

(C) (D)

Q17. The pH of 10^{-3} mol dm^{-3} of an aqueous solution of H_2SO_4 is

(A) (B)

(C) (D)

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

Q. 1123 (Inter Part - I) — Chemistry (Subjective) — Session 2019-21 to 2022-24 — Group (II) — Paper (I)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Define gram atomic mass and gram molecular mass. (ii) Define molecular ion. Give one example. (iii) Mg atom is twice heavier than that of carbon atom. Give reason. (iv) State Graham's Law of diffusion. Write its mathematical form. (v) How the process of respiration obeys the Dalton's law of partial pressure. (vi) Give verification of Boyle's law from kinetic molecular theory of gases. (vii) Why e/m value of cathode rays is just-equal to that of electron. (viii) State Moseley's law. Give its Mathematical expression. (ix) What is orbital? Draw the shape of p-orbital. (x) Define Enthalpy of Atomization. Give one example. (xi) What are spontaneous and non-spontaneous processes. Give one example for each. (xii) State Hess's law of constant heat summation. Write its mathematical form.

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Define sublimation giving two examples. (ii) Give sailent features of a solvent used in process of crystalization. (iii) Describe most safe and reliable method for drying of crystals. (iv) Why melting point and boiling point of halogens increases down the group. (v) Lower alcohols are soluble in water while hydrocarbons are insoluble. Give reason. (vi) Cleavage of crystals is anistropic property. Explain. (vii) Why aquous solution of NH_4Cl is acidic in nature. (viii) Define solubility with two examples. (ix) Why NaCl and KNO_3 are used to lower melting point of ice. (x) Define the term energy of activation. (xi) A catalyst is specific in its action. Justify it. (xii) Rate of reaction decreases with passage of time. Justify it.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) Define electronegativity. How does it vary in the group of periodic table? (ii) Pi (π) bonds are more diffused than sigma bonds. Give the reason. (iii) Define coordinate covalent bond. Give an example. (iv) How can we prepare basic buffers? Give an example. (v) Calculate the pH of 10^{-4} mole dm^{-3} of $Ba(OH)_2$. (vi) Give two applications of common ion effect. (vii) What is standard hydrogen electrode (SHE)? (viii) Give the electrode reactions during the recharging of lead accumulator. (ix) Calculate the oxidation number of Cr in $Cr_2(SO_4)_3$ and $Cr_2O_7^{2-}$

Q. (Marks 8 × 3 = 24) Section ---------- II Note: Attempt any three questions.

Q5. 5. (a) Explain evidence of atoms with the help of diagram.

(b) 250 cm^3 of hydrogen is cooled from 127°C to -27°C by maintaining the pressure constant. Calculate the new volume of the gas at Low temperature.

Q6. 6. (a) Explain molecular solids in detail.

(b) State and explain Hess's law of constant Heat summation with two examples.

Q7. 7. (a) Write down any four properties of cathode rays.

(b) What is the percentage ionization of acetic acid in a solution in which 0.1 Mole of it has been dissolved per dm^3 of the solution.

Q8. 8. (a) Explain paramagnetic nature of oxygen on the basis of MOT.

(b) Describe the construction and working of standard hydrogen electrode (SHE).

Q9. 9. (a) Explain phenol-water system in detail.

(b) Write down any four characteristics of catalyst.

> Note: Board typos kept verbatim: 'sailent', 'crystalization', 'anistropic', 'aquous'.